E-Waste Governance Platform: Build Specification and Procedure

For the UEN008 project at Thapar Institute. Prepared 19 October 2026 from the governance model PDF and the team's workflow description. This is the single reference for the engineer building the website.

1. What the website does

The platform tracks every device from registration to certified recycling, with proof at each step. It has four kinds of users and one rule above all: **every status change needs evidence, and every action is logged.**

Role	Sees and does
Admin / Auditor	The only role that sees the home dashboard: statistics, graphs, user and device data, flagged cases, audit log
User (contributor)	Logs in with registered mobile number or email, sees own devices, tracks stage, chooses payment method, gets notified, downloads certificate
Collection engineer	Registers devices against the user's phone number, wipes data, uploads wipe video, pays, submits dispatch evidence
Factory engineer	Verifies incoming devices and weights, flags errors, records recycling output, issues certificate

Note on the Admin and Auditor: the PDF asks for an auditor who is independent of the operators. Build two sub-roles: **Master Admin** (manages accounts, thresholds, partner list) and **Auditor** (read-only, can export and comment). Both can open the dashboard.

2. End-to-end workflow and status codes

One device has one status at a time. A status can only move forward when the required evidence exists. Build this as a state machine in the backend, not just labels in the interface.

#	Status	Set by	Required before moving on
1	REGISTERED	Collection engineer	Device details entered against user's phone number, intake photo, ownership declaration signed by user
2	DETAILS_VERIFIED	User (via OTP or tap)	User confirms the details shown are correct
3	HOLD (side branch)	Engineer	Locked-and-forgotten, blacklisted IMEI or unclear ownership; device is kept sealed and the owner is informed

#	Status	Set by	Required before moving on
4	WIPING	Collection engineer	Wipe method and tool recorded
5	WIPE_VERIFIED	Collection engineer	Wipe video uploaded, verification result recorded; **user is notified here**
6	PAYMENT_SENT	Collection engineer	Amount, method, transaction reference
7	PAYMENT_ACKNOWLEDGED	User	User taps "I received the payment"
8	READY_FOR_DISPATCH	Collection engineer	Dispatch evidence uploaded (section 5.4) and logged
9	FACTORY_INSPECTION	Factory engineer	Authenticity, weight and quantity checks
10	FLAGGED (side branch)	Factory engineer	Reason code and note; goes to Admin review queue
11	ACCEPTED	Factory engineer	All checks green
12	RECYCLED	Factory engineer	Input and output weights recorded
13	CERTIFIED	System	Certificate generated; **user is notified**

Payment is blocked until WIPE_VERIFIED, and dispatch evidence is blocked until PAYMENT_ACKNOWLEDGED. If the user never acknowledges, send reminders at 3 and 7 days, then let the Admin resolve it manually.

3. Authentication and access control

- **Users:** one-time password (OTP) sent to the registered mobile number or email. No password to store. Session expires after inactivity.
- **Engineers and Admin:** email plus password plus second factor (OTP or authenticator app). Accounts are created by the Master Admin only.
- **Role-based access control enforced on the server**, not by hiding buttons. The dashboard route and every dashboard API return 403 for anyone who is not Admin or Auditor.
- A user can read only their own devices. A collection engineer can edit only devices they registered, and only until the status moves past their step. A factory engineer sees only devices in the dispatch pipeline.
- Rate-limit OTP requests and lock after repeated failures.
- Every create, update, view of evidence and export writes a row to the audit log: who, what, when, from which device.

4. Pages to build

4.1 Admin / Auditor home (dashboard)

This is the landing page after Admin or Auditor login. No other role can reach it.

Summary cards: devices registered, total kg collected, devices wiped, total amount paid, total kg recycled, recycling rate (recycled kg divided by collected kg), flagged devices, open holds.

Graphs:

- Collected versus recycled kg by month (line or bar).
- Pipeline funnel: number of devices at each status.
- Devices by category (phone, laptop, keyboard, charger, battery, board, other).
- Material recovered by fraction (metals, plastics, glass, hazardous, battery).
- Flag rate by reason and by collection engineer.
- Average days spent in each stage.
- Payments by method and total per month.
- Map of dispatch locations from the geotag data (Leaflet with OpenStreetMap).
- Optional energy and carbon panel: kWh per wipe, transport km, estimated avoided emissions, using the method in section 9 of the PDF with the assumptions shown.

Data tables:

- Users with their devices, status, payment and certificate. Mask phone and email by default (show last 4 digits); an unmask action is logged.
- Collected goods table and recycled goods table, filterable by date, category, engineer, status.
- Flag review queue: open a case, see both engineers' entries side by side, resolve with a note.
- Audit log viewer.
- Export to CSV and PDF with the current filters.

4.2 User portal

After OTP login the user sees:

1. **My devices list.** Each card shows brand, model number, UEFI firmware model/version, and condition.
2. **Condition shown as one of:** Working; Locked, password forgotten; Not booting or not opening at all.
3. **Stage tracker.** A horizontal step bar using the status codes (simplified labels: Received, Verified, Data erased, Paid, On the way to recycler, Inspected, Recycled, Certified). The current step is highlighted with its date.
4. **Details confirmation** step: "These details are correct" button.
5. **Payment preference form:** choose UPI ID, or bank account (account number, IFSC, name), and optionally other methods the committee approves. Saved before the engineer pays. Editable until status is PAYMENT_SENT. Show a masked view afterwards.
6. **Wipe proof:** once WIPE_VERIFIED, the user can watch the wipe video.
7. **Acknowledge payment** button after PAYMENT_SENT.

8. **Certificate download** after CERTIFIED.
9. **Notifications** list, mirrored by SMS and email.

Important: a website cannot read a user's UEFI firmware or model number remotely. The **engineer reads these from the device** (system information screen, BIOS/UEFI setup, label, or `dmidecode` / system report) and enters them; the user only sees and confirms them.

4.3 Collection engineer console

1. **Find or create user by phone number.** Verify with an OTP so the right person is linked.
2. **Register device form:** category, brand, model number, serial number, IMEI (phones), UEFI firmware model and version (computers), storage type and size (HDD, SATA SSD, NVMe, eMMC, none), battery present, condition (the three options above), intake photos, notes, ownership declaration checkbox with user confirmation.
3. **Legitimacy checks:** record the result of the IMEI check on the government Know Your Mobile / CEIR service for phones. A blocked or blacklisted IMEI forces the HOLD status and disables payment.
4. **Wipe step:** pick the method from a dropdown tied to the storage type (table in section 6), record tool name and version, start and end time, and measured kWh if available.
5. **Verification:** tool status, plus a sampled recovery test result (pass or fail). A fail returns the device to WIPING or sends the storage to destruction.
6. **Wipe video upload.** See section 5.3 for requirements.
7. **Payment step.** The system suggests an amount from the condition and category (section 7). The engineer confirms, sends payment on the user's chosen method, and records the transaction reference. Status becomes PAYMENT_SENT.
8. **Dispatch evidence icon.** This appears only after PAYMENT_ACKNOWLEDGED. It opens the live capture screen (section 5.4). When saved, the engineer taps "Ready for dispatch" and the device joins the outgoing batch list for the official recycler.

4.4 Factory engineer console

1. **Incoming batch list** with each device's collection-side data and evidence.
2. **Authenticity check:** serial/IMEI, brand and model on the physical device must match the record. Mismatch means flag.
3. **Weight and quantity check:** enter measured weight and counted quantity. The system compares them with the collection engineer's entry and shows the difference.
4. **Flagging rule:** flag automatically when the weight difference exceeds the threshold (suggested starting value: more than 10 percent or 100 g, whichever is larger; the Master Admin can change it) or when quantity or identity does not match. The engineer can also flag manually with a reason code (identity mismatch, weight mismatch, quantity mismatch, evidence missing or unclear, hazard found, other).
5. **Green-flag checklist:** identity matches, weight within threshold, quantity matches, evidence viewed, hazards checked. All green moves the device to ACCEPTED.
6. **Recycling record:** input weight, then output weights by fraction (metals, plastics, glass, circuit boards, battery, hazardous, residue), batch reference, and date. The system calculates recovered percentage per device or batch and updates the dashboard totals.

7. **Complete:** the engineer presses "Mark recycled". The system sets CERTIFIED, generates the certificate and notifies the user.

5. Evidence, media and certificate rules

5.1 Notifications

Send SMS and email (and an in-app message) at: device registered, details awaiting confirmation, wipe verified (the "your device is cleaned" message with a link to the video), payment sent, dispatch, flagged (to Admin, and to the user if the delay is long), recycled, certificate ready.

5.2 Storage

Store videos and photos in private object storage (S3-compatible). Serve through short-lived signed links only. Save a SHA-256 hash of each file at upload so tampering can be detected later. Set a retention period (suggested: keep for the audit period, then delete) and make it configurable.

5.3 Wipe video requirements

The video must show the device identifier (serial or IMEI), the wipe tool's start and finish screen with the method name, and the final verification result. It must **not** show any of the user's personal files, accounts or messages. Give the engineer an on-screen checklist at upload. Reject files that are too short or too large and show a preview before saving.

A video proves a process ran, it does not prove no data remains. That is why the sampled recovery test and independent audit are kept as separate recorded steps.

5.4 Dispatch evidence (the "geo" icon)

Build a capture screen that records **live in the browser** (camera and microphone permission, no gallery upload, to make reuse of old files harder):

- A video of the device on the weighing scale with the display readable, plus the quantity counted aloud or shown.
- A photo of the device.
- Automatic capture of GPS latitude, longitude and accuracy using the browser location permission, plus server timestamp.
- Fields to type the weight in kg and the quantity. The system rejects entry if location is missing or accuracy is poor (suggested limit: worse than 100 m).
- The location is shown on a map in the device record and on the dashboard map.
- Because this needs HTTPS, deploy with a valid certificate from day one.

5.5 Certificate

Generate a PDF with: certificate ID, user name, device brand and model, serial or IMEI (partly masked), date registered, date of data erasure and method, recycler name and registration number, recycling date and batch reference, and a statement that the device was processed through a registered recycler. Add a QR code to a **public verification page** that shows only the certificate ID, device model, status and dates, with no personal details. Sign the PDF or store its hash so it can be checked later.

6. Wipe methods the engineer's dropdown should offer

From the PDF, aligned with NIST SP 800-88 Rev. 2 and IEEE 2883-2022 (confirm against the standards themselves before quoting).

Media	Method	Verification
Hard disk	Overwrite for low sensitivity; purge or destroy for sensitive data	Read-back sampling
SATA SSD	ATA Secure Erase or cryptographic erase	Tool status plus sample recovery test
NVMe SSD	NVMe Sanitize (block or crypto erase)	Controller completion plus sample test
Phone or tablet	Remove accounts and activation lock, encrypt, factory reset, vendor erase where available	Boots to setup screen, sample check
SD card, USB stick	Purge with device method or destroy	Sample check
Network gear, printers	Reset to defaults, sanitise internal storage	Configuration check
Dead or failed media	Physically destroy the storage part only; recycle the rest	Photo and weight log

A plain factory reset is the lowest level (Clear), so the system should warn when it is the only method used on a flash device. A device marked "locked, password forgotten" that cannot be wiped goes to HOLD, and the owner is asked to recover access, or the storage is destroyed with the owner's written consent.

7. Payment rules

The PDF's formula: offer = max(S, a × R) − H, with a floor and a cap, where S is scrap value, R is expected resale value, a is the share passed to the user, and H is handling cost.

- Store the rates in an admin-editable table (category by condition: working, locked, not booting), filled from the local price survey in the PDF (at least ten scrap dealers and two refurbishers).
- Cap payments per person and per day, and require a verified payout account.
- Record every payment with a transaction reference and a receipt the user can view.
- Payment gateway: for the pilot, manual UPI or bank transfer with the reference number typed in is the safest start. Automated payouts (for example through a payment gateway's payout product) can be added later after approvals.
- The PDF recommends a no-cash pilot (points or certificates) until legal approval exists. Build payment as a switch: **cash mode** or **points mode**. Confirm with your teacher and the Zero Waste Team before enabling cash.

8. Data model (main tables)

- **users:** id, name, mobile, email, role, status, created_at.
- **payment_preferences:** user_id, method, UPI ID or bank details (encrypted), updated_at.
- **devices:** id, user_id, category, brand, model_no, uefi_model, uefi_version, serial_no, imei, storage_type, battery_flag, condition, status, registered_by, created_at.
- **ownership_declarations:** device_id, accepted_at, method.
- **wipe_records:** device_id, method, tool, started_at, ended_at, verification_result, recovery_test_result, kwh, video_file_id, engineer_id.
- **payments:** device_id, amount, method, reference, sent_at, acknowledged_at.
- **dispatch_records:** device_id, weight_kg, quantity, lat, lng, accuracy_m, photo_file_id, video_file_id, captured_at, batch_id.
- **factory_checks:** device_id, authentic_flag, weight_kg, quantity, weight_diff_pct, result, factory_engineer_id, checked_at.
- **flags:** device_id, reason_code, note, raised_by, status, resolved_by, resolved_at.
- **recycling_records:** device_id or batch_id, input_kg, outputs by fraction, recycler_id, recycled_at.
- **certificates:** id, device_id, issued_at, pdf_file_id, hash, public_token.
- **files:** id, storage_key, sha256, type, uploaded_by, uploaded_at.
- **notifications:** user_id, channel, message, sent_at, status.
- **settings:** weight threshold, rates table, caps, retention, payment mode.
- **audit_log:** actor_id, action, entity, entity_id, time, ip.

9. Suggested technology

Any stack the team knows is fine. A practical choice:

- Front end: React or Next.js with Tailwind, charts with Recharts or Chart.js, maps with Leaflet.
- Back end: Node.js (Express or NestJS) or Python (Django), REST API with role checks.
- Database: PostgreSQL.
- Files: S3-compatible private bucket.
- OTP and alerts: an SMS provider approved in India (DLT-registered templates) and an email service.
- PDF and QR: a server-side PDF library and a QR library.
- Hosting: HTTPS only, daily database backups, separate staging and production.

10. Security and privacy (DPDP Act duties)

- Collect only what is needed. Do not put owner details in public pages or in the audit certificate record beyond what is required.
- Encrypt data in transit and at rest; encrypt bank and UPI details separately.
- Keep access logs, set a retention schedule, and have a breach-response plan.
- Get legal advice on whether the hub acts as a data fiduciary or processor. The PDF notes that serious penalties apply for security failures.
- Dispatch and factory records must reconcile: the PDF's traceability fixes (weighbridge slips, manifests, random audits) are covered by the weight comparison, geotag evidence and flag queue above.

- Only partner with recyclers registered under the E-Waste (Management) Rules 2022, and store the recycler's registration number in the system.

11. Build order

Step	Work	Done when
1	Database, roles, OTP and staff login, audit log	Each role can log in and sees only its pages
2	Device registration and user confirmation	Engineer registers, user sees and confirms
3	Status engine and stage tracker	Invalid status jumps are rejected
4	Wipe record, video upload, notifications	User is informed at WIPE_VERIFIED
5	Payment preference, payment record, acknowledgement	Payment blocked until wipe verified
6	Live geotagged capture and dispatch list	Evidence unlocks only after acknowledgement
7	Factory checks, flagging, recycling record	Flag fires on threshold breach
8	Certificate PDF and public verification	QR opens a page without personal data
9	Admin dashboard, tables, exports	Totals match the raw records
10	Security review, load test, pilot run	Test scenarios below pass

12. Acceptance tests

- A user cannot see another user's device or open the dashboard URL.
- Payment cannot be sent before wipe verification; dispatch cannot be saved before acknowledgement.
- A device with a blacklisted IMEI cannot be paid and shows HOLD.
- A weight difference above the threshold creates a flag automatically.
- Location off or poor accuracy blocks dispatch evidence.
- The certificate verification page shows no name, phone or email.
- Dashboard totals (collected kg, recycled kg, payments) equal the sum of underlying records.
- Every status change appears in the audit log.

13. Points to confirm before launch

- Permission from your teacher and the Zero Waste Team, and legal advice for cash payment and data handling.
- Written agreement with a registered recycler (and refurbisher if you add a reuse path).
- The weight tolerance, payment caps and rate table values.
- Whether the unregistered student hub may hold devices at all, or must act only as a collection point feeding registered partners, as the PDF advises.

14. Suggested additions from the PDF (optional, after the core works)

- **Reuse branch:** a Grade A/B device can go to a registered refurbisher instead of recycling, with its own certificate wording.
- **Hazard step:** battery check at intake and a sealed battery stream.
- **Public scorecard** with anonymous totals, and hostel or department leaderboard to support awareness.
- **Intake photo classifier** suggesting the category, as a demonstration only.